

Development Automation **& Linux Lab**

Experiment 2

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Batch: 03

Semester IV

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UNIVERSITY OF PETROLEUM & ENERGY STUDIES

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2022-23

EXPERIMENT-2

Introduction to Bash Scripts

1. Write a script to enter marks by user in the terminal. If the entered mark is greater than 35 then output pass, otherwise output fail.

```
#!/bin/bash

echo "Enter Marks"
read MARKS
if [ $MARKS -gt 35 ];
then
    echo "pass"
else
    echo "fail"
fi
```

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$ ./ques1.sh
Enter Marks
34
fail
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$ ./ques1.sh
Enter Marks
35
fail
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$ ./ques1.sh
Enter Marks
36
pass
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$
```

2. Write a script to enter marks by user in the terminal. If marks are greater than 80 output distinction, if marks between 60 and 80 then output first division. If marks are less than 60 and greater than or equal to 35 then output pass, otherwise output fail.

```
#!/bin/bash

echo "Enter Marks"
read MARKS
if [ $MARKS -gt 80 ];
then
    echo "distinction"
elif [ $MARKS -le 80 -a $MARKS -ge 60 ];
then
    echo "first division"
elif [ $MARKS -lt 60 -a $MARKS -ge 35 ];
then
    echo "pass"
else
    echo "fail"
fi
```

```

ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$ ./ques2.sh
Enter Marks
81
distinction
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$ ./ques2.sh
Enter Marks
80
first division
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$ ./ques2.sh
Enter Marks
75
first division
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$ ./ques2.sh
Enter Marks
60
first division
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$ ./ques2.sh
Enter Marks
40
pass
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$ ./ques2.sh
Enter Marks
34
fail
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$

```

3. Write a script to print numbers from 1 to 10 using while loop.

```

#!/bin/bash

num=1

while [ $num -le 10 ]; do
    echo $num
    num=$((num+1))
done

ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$ ./ques3.sh
1
2
3
4
5
6
7
8
9
10

```

4. Write a script to find the average of first N natural numbers, where N is entered by the user at the terminal.

```

#!/bin/bash

echo "Enter Number"
read num
cpnum=num
avg=0
i=0
while [ $i -le $num ]; do
    avg=$((avg+i))
    i=$((i+1))
done
echo $avg
avg=$((avg/num))
echo "Average of first $num natural numbers is $avg"

# Bash doesn't support floating-point arithmetic

ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$ ./ques4.sh
Enter Number
4
10
Average of first 4 natural numbers is 2
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$

```

5. Write a script to find the average of N numbers of an array (of size N). The values of the array are entered by the user at the terminal. (Use while loop)

```
#!/bin/bash

echo "Enter Size of Array"
read size
array=()
i=0
avg=0

echo "Enter Elements in Array"

while [ $i -lt $size ]; do
    read value
    array[i]=$value
    i=$((i+1))
    avg=$((avg+value))
done

avg=$((avg/size))

echo "Average = $avg"
```

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$ ./ques5.sh
Enter Size of Array
5
Enter Elements in Array
1
2
3
4
7
Average = 3
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts$
```